



ICPC2T 2027



IEEE

Student Branch
NIT Raipur



IEEE
MADHYA PRADESH SECTION

2027 IEEE International Conference on

Power, Control and Computing Technologies (ICPC2T 2027)

NIT Raipur, Chhattisgarh, India | 3–5 March 2027

SPECIAL SESSION (SS1) ON

Intelligent Materials and Electrochemical Technologies for Power, Hydrogen, Energy Storage and Sensing Applications

LIST OF TOPICS

- Nanomaterials, graphene, hybrid composites, sustainable materials
- MXenes, graphene, carbon materials, metal oxides, sulfides and phosphides
- Green hydrogen, water splitting, electrocatalysis, photocatalysis, sensors
- Batteries, state-of-charge, state-of-health supercapacitors, hybrid storage, electrode engineering, intelligent battery and supercapacitor management systems
- AI/ML, digital twins, predictive modeling, intelligent control
- Smart sensors, embedded systems and wireless monitoring, fault detection
- Integration of storage, hydrogen and sensor platforms with microgrids, Energy harvesting
- Electrochemical impedance spectroscopy, signal processing and diagnostics
- Microgrids, power electronics, energy management, renewable integration
- Thermal management, safety and reliability, data-driven prediction of efficiency
- Renewable-energy smoothing, peak management and optimal power sharing
- Additive manufacturing, laser processing, in situ characterization, prototyping
- Flexible, wearable and miniaturized electrochemical devices
- Photoelectrochemical and photocatalytic hydrogen-generation technologies
- Recycling, lifecycle assessment, circular economy, techno-economic analysis, standard
- Prototypes, demonstrators and technology-readiness advancement
- Standards, cybersecurity and communication for smart energy devices

SPECIAL SESSION CHAIRS



Dr. Yogesh Kumar K

Alliance University,
Bengaluru



Dr. Jayaraman Theerthagiri

Gyeongsang National University,
Jinju, Republic of Korea



Dr. Arun Varghese

Alliance University,
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SUBMISSION DETAILS

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After login, select Special Session (SS1): “Intelligent Materials and Electrochemical Technologies for Power, Hydrogen, Energy Storage and Sensing Applications”

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